

William Dong

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EDUCATION

Carnegie Mellon University

B.S. Mechanical Engineering, Freshman

Expected May 2030

Great Valley High School

Senior · Malvern, PA · GPA 4.0 / 4.0

2022 – 2026

Relevant coursework: AP Physics C (Mechanics, E&M), AP Calculus BC, AP CSA

EXPERIENCE

Student Researcher

Carnegie Mellon University · Robotics Institute · Biorobotics Lab

2026 – Present

- Selected for the Biorobotics Lab within Carnegie Mellon's top-ranked Robotics Institute, a world leader in bio-inspired and snake robotics.
- Currently leading a novel locomotion project as a freshman, the only undergraduate in the lab to lead one: a robot that fires an anchor onto a structure and pulls itself toward it, developed from zero prior work.
- Also developing and testing experimental hardware for a bio-inspired snake robot.

Accelerator-Backed Builder

Scotty Labs Foundry · Carnegie Mellon University

2026 – Present

- Selected by application to the founding cohort; building ground UAS systems.

Founder & CEO

Formula Theory & Automotive Design (FTAD) · 501(c)(3) Nonprofit

2024 – Present

- Founded an F1-inspired robotics competition nonprofit, securing \$55,000+ in funding through targeted sponsorships and grant acquisition.
- Conducted extensive market research and obtained international product validation, establishing FTAD's competition framework across multiple countries.
- Engineered competition vehicle designs in Onshape and Fusion 360, applying CFD simulation and topology optimization to maximize aerodynamic and structural performance.

MechE Intern, UGV Defense Company

Raptor Defense Company

Summer 2026

- Led SCORPION, an expendable payload vehicle, as lead and sole mechanical engineer, owning the full mechanical design from concept to working prototype.
- Designed the complete assembly in Onshape in 6 days, then manufactured and assembled it in 3.
- Delivered on spec: exceeded 50 mph (conservative estimate), used under 50% of the R&D budget, and passed the preliminary test matrix.
- Built a Portable Control Unit for the electronics team in SolidWorks.

Design Lead & CAD Specialist

VEX Robotics · World-Qualified Competitive Team

2024 – 2026

- Led all mechanical design and CAD work on a team that qualified for the VEX World Championship in the 2024 season.

LEADERSHIP & ACTIVITIES

Competitive Boxer

USA Boxing

2023 – Present

- Two-time PA State Champion (2024 & 2025).
- Volunteer boxing coach, mentoring athletes ages 12–34 in technique, discipline, and sportsmanship.

Future Business Leaders of America (FBLA)

Web Development · National Qualifier · 4th in Pennsylvania

2025

- Learned web-dev logic in 4 months and built the competition project in a team of 3.

Co-Founder

NextGen Business & Law Forum · 501(c)(3) Nonprofit

2022 – 2024

- Organized entrepreneurship competitions for 500+ students across the United States.
- Oversaw event logistics, curriculum development, and partner outreach.

SKILLS

CAD & Design: Onshape, Fusion 360, SolidWorks, CFD Simulation, Topology Optimization

Programming: Python, Java, C++

Engineering: Kinematic Systems, Structural Analysis, Aerodynamic Simulation, Rapid Prototyping, UGV Systems

INTERESTS

Kinematics engineering, automotive engineering & racing technology, unmanned systems, defense technology, entrepreneurship, and combat sports.